



DAFFODIL INTERNATIONAL UNIVERSITY

Department of Computing and Information System (CIS)

Web Development essentials lab

Lab JS Task in Class

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Class Task-01

Practice Problem -01:

```
let money=1000; let cost=700; let result=money-cost;  
console.log("The Return is:",result);
```

Output:

```
PS C:\Users\AILAB\OneDrive\Desktop\Rashed> node practice_problem_1.js  
The Return is: 300  
PS C:\Users\AILAB\OneDrive\Desktop\Rashed> █
```

Practice Problem -04:

```
var a=isNaN('11'); var a=isNaN(2-10);  
console.log(a);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\practice_4.js"  
false
```

JS Condition Task-1:

```
let burgerPrice=600;  
if(burgerPrice>500){  
  console.log("Free Coke");  
}  
else{  
  console.log("Coke:30tk  
");  
}
```

Output:

```
PS C:\Users\AILAB\OneDrive\Desktop\RAshed> node practice_problem_1.js
The Return is: 300
PS C:\Users\AILAB\OneDrive\Desktop\RAshed> █
```

JS Condition Task-2:

```
let weight = 70; let height =
1.75; let bmi = weight /
(height * height);
console.log("BMI =", bmi); if
(bmi < 18.5) {
console.log("Underweight");

} else if (bmi >= 18.5 && bmi
<= 24.9) {
console.log("Normal");

} else if (bmi >= 25 && bmi
<= 29.9) {
console.log("Overweight");

} else {
console.log("Obese
");
}
}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\BMI_test.js"
BMI = 22.857142857142858
```

JS Condition Task-3:

```
let marks = 85; if
(marks >= 90) {
console.log("Grade
A");

} else if (marks >=
80) {
console.log("Grade
B");

} else if (marks >=
70) {
```

```
console.log("Grade  
C");  
  
} else if (marks >=  
60) {  
console.log("Grade  
D");  
  
} else {  
console.log("Grade  
F");  
  
}
```

Output:

```
PS C:\Users\AILAB\OneDrive\Desktop\RASHed> node grade_calculator.js  
Grade B  
PS C:\Users\AILAB\OneDrive\Desktop\RASHed> █
```

JS Condition Task-4:

```
let myScore  
=85; let  
friendScore =  
65;  
  
if (myScore > 80) {      if  
(friendScore > 80) {  
console.log("Go for a lunch");  
    }  
    else if (friendScore >= 60) {  
console.log("Good luck next time");  
    }  
    else if (friendScore >= 40) {  
        console.log("Keep friend's message unseen");  
  
    }  
    else  
    e {  
        console.log("Block your  
friend");  
    }  
}
```

```
}  
}  
e  
l  
s  
e  
{  
  
    console.log("Go home, sleep and act  
sad"); }  

```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\nested_ifElse.js"  
Good luck next time
```

JS Condition Task-5:

```
let  
num1=20;  
let  
num2=10;  
  
let result;  
  
if(num1>num  
2){  
    result=num1  
    *2;  
  
} else{  
    result=num1+nu  
m2;  
  
}  
  
console.log(result);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\example_5.js"  
40  
  
[Done] exited with code=0 in 0.136 seconds
```

JS Condition Task-5: (ternary)

```
Let
num1=20;
let num2 =
15;
let result = (num1 > num2) ? num1 * 2 : num1
+ num2;
console.log(result);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\Example_5_ternary.js"
40
```

JS Condition Task-6:

```
let age = 2;
let isStudent
= true; let
fare = 800;

  if (age < 10) {
console.log("Free
Ticket");

} else if (isStudent) {
console.log("Fare =", fare *
0.5);

} else if (age >= 60) {
console.log("Fare =", fare *
0.85);

} else {
console.log("Fare =",
fare);

}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\example_6.js"
Free Ticket
```

JS loop Task-1:

```
let sum=0; let  
i=1;  
while(true){  
sum+=i;  
if(sum>=100){  
break;    }  
i++; }  
console.log(sum)
```

; Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\loop1.js"  
105
```

JS Loop Task-2:

```
let arr=[]; for(let i=1;  
i<=40; i++){  
if(i%2==0){  
arr.push(i);  
    } }  
console.log(arr)  
;  
console.log(sum)  
;
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\loop2.js"  
[  
  2,  4,  6,  8, 10, 12, 14,  
 16, 18, 20, 22, 24, 26, 28,  
 30, 32, 34, 36, 38, 40  
]
```

Loop Task-3:

```
let arr=[];  
for(let i=55; i<=85; i++){  
  if(i%2!=0 && i%5!=0){  
    arr.push(i);  
  }  
}  
console.log(arr)  
;
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\loop3.js"  
[  
  57, 59, 61, 63, 67,  
  69, 71, 73, 77, 79,  
  81, 83  
]
```

JS Array Task-1:

```
let fruits = ["Apple", "Banana", "Orange", "Mango",  
"Lichi"]; console.log(fruits[2]); fruits[2] = "Jambura";  
console.log(fruits);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\array1.js"  
Orange  
[ 'Apple', 'Banana', 'Jambura', 'Mango', 'Lichi' ]
```

Array Task-2:

```
let tourist = ["Cox's Bazar", "Bandarban",  
"Sylhet"]; tourist.push("Rangamati");  
tourist.push("Sajek"); tourist.push("Kuakata");  
tourist.pop(); console.log(tourist);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\array2.js"  
[ "Cox's Bazar", 'Bandarban', 'Sylhet', 'Rangamati', 'Sajek' ]
```


JS Array Task-3:

```
let books=["Math", "English", "JavaScript", "Physics"];
if (books.includes("JavaScript")) {
  console.log("JavaScript book found");

} else {
  console.log("Not found");
}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\array3.js"
JavaScript book found
```

JS Array Task-4:

```
let a = [1,2,3];
let b = "Hello";
let c = 100;

console.log(Array.isArray(a)); console.log(Array.isArray(b));
console.log(Array.isArray(c));
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\array4.js"
true
false
false
```

JS Array Task-5:

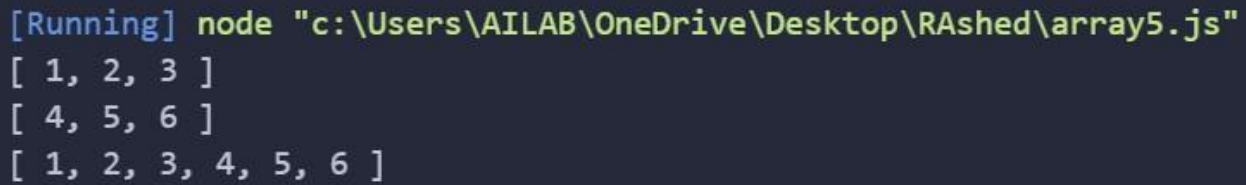
```
let arr1 = [1,2,3];
let arr2 = [4,5,6];

let arr3 =
arr1.concat(arr2);

console.log(arr1
);
```

```
console.log(arr2  
);  
console.log(arr3  
);
```

Output:



```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\array5.js"  
[ 1, 2, 3 ]  
[ 4, 5, 6 ]  
[ 1, 2, 3, 4, 5, 6 ]
```

JS Object Task 1:

```
const colors = {    red: "#ff0000",    green: "#00ff00",  
blue: "#0000ff",  
    "golden rod": "#daa520"  
}; console.log(colors["golden rod"]);
```

Output:



```
velopment Essential  
#daa520
```

JS Object Task 2:

```
const car = {      make: "Toyota",      model:
"Corolla",      year: 2020,

      passenger_capacity: 5
}; console.log(car);
```

Output:

```
velopment Essentials\Java\passenger.js"
{ make: 'Toyota', model: 'Corolla', year: 2020, passenger_capacity: 5 }
```

JS Object Task 3:

```
const student = {      name: "Hamim Sakep",      id:
5421,      physics: {      subject: "HSC Physics",
author: "Shahjahan Tapan",      marks: 30      }
}; console.log(student.physics.marks);
```

Output:

```
PS C:\Users\HP\Desktop\CIS_3rd_Semester\We
velopment Essentials\Java\marks.js"
30
PS C:\Users\HP\Desktop\CIS_3rd_Semester\We
```

JS Object Task 4:

```
let student = {      name: 'Ariana Grande',      age: 21,
city: 'Gaibandha',      isStudent: true }; let count =
Object.keys(student).length;

console.log(count);
```

Output:

```
PS C:\Users\HP\Desktop\CIS_3rd
velopment Essentials\Java\cour
4
PS C:\Users\HP\Desktop\CIS_3rd
```

JS Object Task 5:

```
let myObject = {      name: 'John Doe',      age: 25,  
city: 'Example City',      isStudent: true  
}; for (let key in myObject) {      console.log("key:",  
key, "| type:", typeof myObject[key]);}
```

Output:

```
key: name | type: string  
key: age | type: number  
key: city | type: string  
key: isStudent | type: boolean
```

JS String Task-1:

```
let str = "banana";  
let count = 0;  
for(let ch of str){  
  if(ch==='a'){  
    count++;  
  }  
}  
console.log(count);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\string1.js"  
3
```

JS String Task-2:

```
let str = "bAnanA"; let  
count = 0; for(let ch of  
str){      if(ch==='a' ||
```

```
ch==='A') {  
    count++;  
}  
}  
console.log(count);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\string2.js"  
3
```

JS String Task-3:

```
const prompt=require("prompt-sync")(); let str=prompt("Enter a  
string: "); if(str.includes('a') &&  
str.includes('e')&&str.includes('i')&&str.includes('o')&&str.in  
cludes('u')){ console.log("All vawels here");  
} else{ console.log("Doesn't contain  
all vowels"); }
```

Output:

```
> node "I:\Programming\All Coding\Java Script\basic\string3.js"  
Enter a string: education  
All vawels here  
PS I:\Programming\All Coding>
```

JS String Task-4:

```
const prompt=require("prompt-sync")();  
let str=prompt("Enter a string: ");  
  
str=str.replaceAll('x','y')  
);  
str=str.replaceAll('X','Y')  
);
```

```
console.log(str);
```

Output:

```
PS I:\Programming\All Coding> node "I:\Programming\All Coding\Java Script\basic\string4.js"  
Enter a string: Xerox xylophone  
Yeroy yylophone  
PS I:\Programming\All Coding>
```

JS String Task-5:

```
const prompt=require("prompt-sync")();
let str=prompt("Enter a string: ");

let word=str.split(" "); for(let
i=0;i<word.length;i++){
word[i]=word[i][0].toUpperCase()+word[i].slice(1);
} let
result=word.join(" ");
console.log(result);
```

Output:

```
> node "i:\Programming\All Coding\Java Script\basic\string5.js"
Enter a string: this is rashed
This Is Rashed
PS I:\Programming\All Coding> |
```

Class Task-2

Task-01:

```
const heights2 = [167, 190, 120, 165, 137];
let lowest = heights2[0];
for(let i = 1; i < heights2.length; i++)
{
    if(heights2[i] < lowest)
    {
        lowest = heights2[i];
    }
}
console.log(lowest);
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\task1.js"
120

[Done] exited with code=0 in 0.299 seconds
```

Task -02:

```
const heights2 = ['rahim', 'robin', 'rafi', 'ron', 'rashed'];
```

```
let smallestName = heights2[0];
for(let i = 1; i < heights2.length; i++)
{
    if(heights2[i].length < smallestName.length)
    {
        smallestName = heights2[i];
    }
}
console.log(smallestName);
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\task2.js"
ron

[Done] exited with code=0 in 0.24 seconds
```

Task-3:

```
function calculateElectronicsBudget(laptop, tablet, mobile){
    let laptopPrice = 35000;
    let tabletPrice = 15000;
    let mobilePrice = 20000;

    let total = (laptop * laptopPrice) +
                (tablet * tabletPrice) +
                (mobile * mobilePrice);

    return total;
}
console.log(calculateElectronicsBudget(1, 1, 3));
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\task3.js"
110000

[Done] exited with code=0 in 0.117 seconds
```

Task-4:

```
const phones = [
    { model: "PhoneA", brand: "Iphone", price: 95000 },
```

```
    { model: "PhoneB", brand: "Samsung", price: 40000 },
    { model: "PhoneC", brand: "Oppo", price: 26000 },
    { model: "PhoneD", brand: "Nokia", price: 35000 },
    { model: "PhoneE", brand: "Iphone", price: 105000 },
    { model: "PhoneF", brand: "HTC", price: 48000 }
  ];
function findAveragePhonePrice(phones){
  let total = 0;

  for(let i = 0; i < phones.length; i++)
  {
    total = total + phones[i].price;
  }
  let average = total / phones.length;

  return average;
}
console.log(findAveragePhonePrice(phones));
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\task4.js"
58166.666666666664

[Done] exited with code=0 in 0.213 seconds
```

Task-5:

```
const employees = [
  {name: "Rashed", experience: 5, starting: 20000, increment:
5000},
  {name: "Amit", experience: 3, starting: 15000, increment:
7000},
  {name: "Pantho", experience: 9, starting: 30000, increment:
1000},
  {name: "Tasrif", experience: 0, starting: 29000, increment:
4000}
];
let totalSalary = 0;
for(let i = 0; i < employees.length; i++)
{
  let salary = employees[i].starting +
```



```
        (employees[i].experience *  
employees[i].increment);
```

```
    totalSalary = totalSalary + salary;  
}  
console.log(totalSalary);
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\task5.js"  
149000  
  
[Done] exited with code=0 in 0.263 seconds
```

Problem-1:

```
function totalFine(fare)  
{  
    if(typeof fare !== "number" || fare <= 0)  
    {  
        return "Invalid";  
    }  
  
    let fine = fare + (fare * 20 / 100) + 30;  
  
    return fine;  
}  
let fare=200;  
// let fare=Number(prompt("Enter fare:"));  
console.log(totalFine(fare));
```

Output:

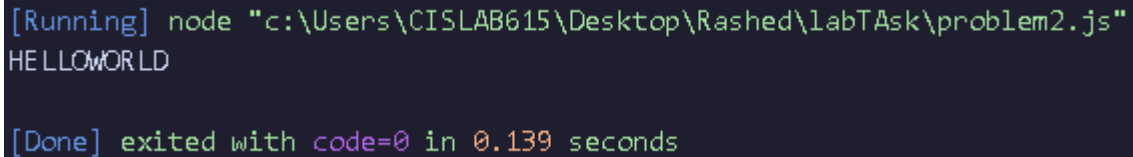
```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\problem1.js"  
270  
  
[Done] exited with code=0 in 0.11 seconds
```

Problem-2:

```
function onlyCharacter(str){  
    if(typeof str !== "string")  
    {
```

```
        return "Invalid";
    }
    let result = "";
    for(let i = 0; i < str.length; i++){
        if(str[i] !== " ")
        {
            result = result + str[i];
        }
    }
    return result.toUpperCase();
}
console.log(onlyCharacter(" h e llo wor ld"));
```

Output:



```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\problem2.js"
HELLOWORLD

[Done] exited with code=0 in 0.139 seconds
```

Problem-3:

```
function bestTeam(team1, team2) {
    if (typeof team1 !== "object" || typeof team2 !== "object") {
        return "Invalid";
    }

    let team1Violation = team1.foul + team1.cardY + team1.cardR;
    let team2Violation = team2.foul + team2.cardY + team2.cardR;

    if (team1Violation < team2Violation) {
        return team1.name;
    }
    else if (team2Violation < team1Violation) {
        return team2.name;
    }
    else {
        return "Tie";
    }
}
let team1 = {
    name: prompt("Enter Team 1 name:"),
```

```
foul: Number(prompt("Enter Team 1 foul:")),  
cardY: Number(prompt("Enter Team 1 yellow card:")),  
cardR: Number(prompt("Enter Team 1 red card:"))  
};  
let team2 = {  
  name: prompt("Enter Team 2 name:"),  
  foul: Number(prompt("Enter Team 2 foul:")),  
  cardY: Number(prompt("Enter Team 2 yellow card:")),  
  cardR: Number(prompt("Enter Team 2 red card:"))  
};  
console.log(bestTeam(team1, team2));
```

Output:

```
Start small. Ship something.  
Enter Team 1 name:Brazil  
Enter Team 1 foul:9  
Enter Team 1 yellow card:10  
Enter Team 1 red card:5  
Enter Team 2 name:Argewntina  
Enter Team 2 foul:20  
Enter Team 2 yellow card:12  
Enter Team 2 red card:0  
Brazil  
  
=== Code Execution Successful ===
```

Problem-4:

```
function isSame(arr1, arr2)  
{  
  if(!Array.isArray(arr1) || !Array.isArray(arr2))  
  {  
    return "Invalid";  
  }  
  
  if(arr1.length !== arr2.length)  
  {  
    return false;  
  }  
  for(let i = 0; i < arr1.length; i++)  
  {  
    if(arr1[i] !== arr2[i])
```

```
        {
            return false;
        }
    }

    return true;
}

console.log(isSame([10, 22, 32],243));
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\problem4.js"
Invalid

[Done] exited with code=0 in 0.402 seconds
```

Problem-10:

```
function resultReport(marks) {
    if(!Array.isArray(marks)) {
        return "Invalid";
    }
    let total = 0;
    let pass = 0;
    let fail = 0;
    for(let i = 0; i < marks.length; i++){
        total = total + marks[i];
        if(marks[i] >= 40){
            pass++;
        }
        else{
            fail++;
        }
    }
    let finalScore = 0;
    if(marks.length > 0) {
        finalScore = Math.round(total / marks.length);
    }
    return {
        finalScore: finalScore,
        pass: pass,
        fail: fail
    };
}
```

```
console.log(resultReport([98, 87, 67, 91, 92, 33, 87]));
```

Output:

```
[Running] node "c:\Users\CISLAB615\Desktop\Rashed\labTask\problem5.js"  
{ finalScore: 79, pass: 6, fail: 1 }  
  
[Done] exited with code=0 in 1.559 seconds
```

End